



Research paper

Synthesis and application of indole esters derivatives containing acrylamide group as antifouling agents

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ARTICLE INFO

Keywords:

Indole derivatives
Antibacterial activity
Algal inhibiting performance
Protein resistance
Antifouling

ABSTRACT

Two new indole ester derivatives containing acrylamide group were synthesized by Michael addition reaction and Friedel-Crafts reaction based on 5-bromo-1*H*-indole. The synthesized compounds were characterized by ¹H Nuclear Magnetic Resonance Spectroscopy (¹H NMR) and Fourier Transform Infrared spectroscopy (FTIR). In the step of the Friedel-Crafts reaction, the optimum process conditions were determined as that the catalyst was AlCl₃, the reaction time was 3 days and the reaction temperature was 25 °C. The results of minimal inhibitory concentration (MIC) of synthesized compounds showed that the two synthesized compounds have better antibacterial activity than reference. Algal inhibiting experiment shows that indole derivatives have better algae inhibition performance. Anti-protein adsorption experiment of antifouling paints with indole derivative and chlorothalonil as antifouling agents proves that indole derivatives exhibit better protein resistance behaviors. The antifouling paints using indole derivatives as antifouling agents have excellent antifouling properties after 24 months marine field test.

1. Introduction

Marine biofouling causes huge economic losses and serious problems to maritime industries [1–3]. The application of an antifouling paint on ships is an effective and convenient method to address potential issues caused by organism adhesion to ships [4,5]. The antifouling agents play pivotal role in the process of marine antifouling coatings as key components. The earliest antifouling agents are copper, lead and their compounds, and later arsenic, mercury and their compounds. Consecutively, tributyltin (TBT) is widely used in the 1970 s. However, they are banned due to their serious environmental pollution and ecological damage since January 1, 2003. Most of the antifouling agents contained in existing commercialized antifouling paints also lead to various levels of pollution in the marine environment, such as chlorothalonil and cuprous oxide. These make the development of environmentally friendly antifouling agents imminent, and many new antifouling agents have been developed [6–8].

During the development of environmentally friendly antifouling agents, searching for substances with antifouling activity from the natural environment, especially secondary metabolites of marine life, is one of the ideal ways to solve the environmental pollution of toxic

antifouling agents such as TBT. Therefore, active antifouling natural products used as antifouling agents have been extracted from marine sessile organisms [9–13]. Since its first synthesis in 1866, indole was found in many natural products and marine natural products [14–16]. In the last few years, it has been reported that indole myriad derivatives and its biosisters have a wide spectrum of bioactivities ranging from antimicrobial to anticancer activities [15–19]. The indole compounds extracted from marine organisms used widely in marine antifouling not only have highly antifouling performance but also are harmless to the marine environment [20,21]. Although the natural product antifouling agents have many advantages such as harmlessness, high efficiency, etc., but the extraction is difficult and the yield is low. Therefore, the synthesis of new low-toxic antifouling agents is an effective and important method. Compared with other natural antifouling agents, indole compounds not only have high antifouling activity and the growth-inhibiting effect on environmental plants, insects and microorganisms, but also are easy to synthesize due to their simple structure. With such characteristics, a large number of novel indole derivatives have been synthesized. Chitra, G et al. antifungal heterocyclic hydrogels using citric acid as a raw material, and synthesized biopolymer hydrogels with properties of anti-oxidation and antifungal based on indole-3-acetic acid (IAA) by

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<https://doi.org/10.1016/j.cplett.2021.138994>

Received 9 July 2021; Received in revised form 18 August 2021; Accepted 19 August 2021

Available online 23 August 2021

0009-2614/© 2021 Published by Elsevier B.V.

polycondensation reaction. The anti-fungal activity of hydrogel samples against *Aspergillus fumigatus*, *Rhizopus* and *Candida albicans* at different concentrations was tested. The results showed that the synthetic indole polymer hydrogels have good antibacterial activity [22]. A series of novel indole dihydronaphthoheterocyclic analogues were synthesized via catalyzed indole by Lewis acid adding to arylene/heteroaryl ketones. The significant antibacterial properties of synthesized compounds against *Escherichia coli* (*E. coli*) and gold *Staphylococcus aureus* (*S. aureus*) was confirmed by antibacterial test [23]. A new isatin-indole molecular hybrids 5a-n were synthesized by molecular hybridization technique. The results of diameter of inhibition zone (DIZ) and MIC against gram-negative bacteria and gram-positive bacteria indicated that the prepared compounds 5a-n have different degrees of antibacterial activity in vitro [24]. Ghorai J et al synthesized cyclopropane-fused indolines in good yield a metal-free directed C2, C3-cyclopropanation. This method works well with a wide range of differently substituted α -diazo esters as well as indole derivatives [25]. Zhou G D et al prepared divergent indole-2-carboxylic acid derivatives including esters, amides and anhydrides by ligand-free copper-catalyzed Ullmann coupling reaction. In the experiment, the yield of synthetic compound was good with aldehydes as raw materials [26].

In order to develop new type of indole antifouling agents with high efficiency and low toxicity, indole esters derivatives contain acrylamide group were synthesized by Michael addition reaction and Friedel-Crafts reaction. The specific technical route of synthetic compounds is as follows (see Scheme 1):

Meanwhile, in the step of the Friedel-Crafts reaction, the influencing factors such as catalyst type, reaction time and reaction temperature were explored to optimize the reaction conditions and improve the yield. The synthesized compounds were identified by FTIR and ^1H NMR. The antibacterial properties and algae inhibition performance of the synthesized compounds were inspected. At the same time, marine antifouling paints using the compounds as antifouling agents were prepared and their antifouling properties were detected.

2. Experimental

2.1. Materials

5-bromo-1*H*-indole was purchased from Chongqing Yingsikai Chemical Co., Ltd. (China). Dimethyl β -dithiodipropionate, diethyl β -dithiodipropionate and *N*-methylol acrylamide were obtained from Shanghai Meixing Co., Ltd. (China). *S. aureus* and *E. coli* inoculum were provided by the microbiology lab of the College of Marine Life Science, Ocean University of China (China). Three marines algal *Nitzschia closterium* (*N. Closterium*), *Chlorella vulgaris* (*C. vulgaris*) and *Skeletonema costatum* (*S. costatum*) which were provided by Key Laboratory of Marine

Chemistry Theory and Technology, Ministry of Education, Ocean University of China (China). Albumin from bovine serum (BSA) were obtained from Shanghai Aladdin Bio-Chem Technology Co., Ltd. (China). Coomassie brilliant blue G250 was purchased from Sigma-Aldrich (USA). All other reagents were purchased from Tianjing Damao Chemical Reagent Factory (China). All the chemicals were of analytical grade.

2.2. Synthesis of compounds

2.2.1. Syntheses of compounds 1a and 1b

Take 1a target compound as examples. 9.8 g (0.05 mol) of 5-bromo-1*H*-indole and 11.9 g (0.05 mol) of dimethyl β -dithiodipropionate were weighed and dissolved in 50 mL of anhydrous tetrahydrofuran (THF) Separately. 50 mL anhydrous THF and 3.6 g (0.09 mol) NaH (Content is 60%) was added in a three-necked flask (250 mL) equipped with a stirrer and thermometer, then the THF solution containing 5-bromo-1*H*-indole was dropwise in the flask keeping the reaction temperature at 0 °C. The reaction was performed for 30 min after dropping, after that, the THF solution containing dimethyl β -dithiodipropionate was dropwise in the above mixture with the reaction temperature remaining at 0 °C. The reaction was performed for 3 h at room temperature. Subsequently, 50 mL of saturated NH_4Cl solution was added to the reactant, then placed for a while, the oil layer was taken out. The solvent was evaporated from oil layer, the left substance was washed with water, after drying for 12 h on MgSO_4 , red-brown oil was obtained. With toluene as eluent, the crude product was separated by column chromatography to give a yellow product 1a. The yield of 1a was 62.2%, the yield of 1b was 65.3%.

2.2.2. Syntheses of compounds 2a and 2b

Take 2a target compound as examples. 14.1 g (0.05 mol) of methyl 3-(5-bromo-1-indolyl)propionate, 10.1 g (0.1 mol) of *N*-methylol acrylamide, 100 mL of absolute ethanol, 4.0 g of aluminum trichloride were placed in a three-necked flask (500 mL) which is fitted with a thermometer, an electric agitator and a water cooled condenser. The reaction was performed for 36 h under 25 °C in water bath. After the reaction, ethanol was removed by rotary evaporation, then concentrate was poured into a dilute hydrochloric acid solution (concentrated hydrochloric acid to water in a volume ratio of 1:3) and stirred, the catalyst aluminum trichloride was removed. The brick red crude product was obtained after it was filtered off and dried. With volume ratio of 1:2 of ethyl acetate-chloroform as eluent, the crude product was separated by column chromatography, a white solid 2a was obtained. The yield of 2a was 80.7%, with melting point at 138.5 to 139.5 °C; the yield of 2b was 83.4%, with melting point at 141.8 to 143.1 °C.

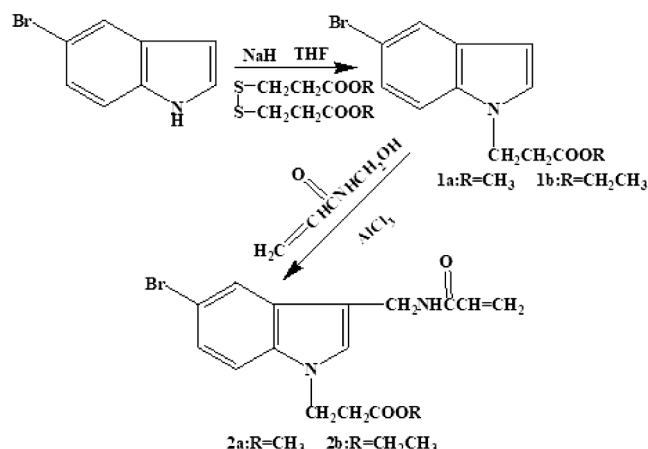
2.3. Preparation of marine antifouling paints

All the components were added in a glass jar with a sealed lid. Then a certain amount of steel balls was putting in and sealed, using coating rapid dispersion testing machine to disperse for 20–30 min, the desired antifouling paint was obtained. The coating composition is shown in Table 1.

2.4. Characterization methods

2.4.1. FTIR

1–2 mg sample and dry KBr were grinded into a powder (about 100



Scheme 1. Synthetic routes of compounds.

Table 1
Composition of marine antifouling coatings.

Antifouling agents type	Rein	Antifouling agents	Copper (I) oxide	pigments and fillers	solvent
chlorothalonil	30	0	30	25	15
2a	30	10	20	25	15
2b	30	10	25	25	15

mg with a particle size of 200 mesh) in an agate mortar, then mixed and loaded into a mold, compressed into thin slices on a tablet press. FTIR spectra were recorded in the infrared absorption range of 400–4000 cm^{-1} used AVATAR-360 Fourier Transform Infrared Spectrometer. The background spectrum was collected firstly, and then the **sample was scanned**.

2.4.2. ^1H NMR

The synthesized compound was purified and dried, and was dissolved with CDCl_3 as solvent. The sample solution was filtered to ensure that the prepared sample solution is evenly mixed, without bubbles or solids. ^1H NMR spectra were recorded with AVANCE III (600 MHz) instrument using TMS as an internal standard, and chemical shifts were indicated in parts per million (δ).

2.5. Cultivation of bacteria

Beef extract Peptone liquid medium: beef extract 1.5 g, peptone 5 g, sodium chloride 2.5 g and water 500 mL were mixed and pH was 7.0 to 7.2. The mixture was sterilized at 121 $^\circ\text{C}$ for 20 min and placed in a refrigerator. Beef paste peptone solid medium: nutrient agar 33 g and water 1000 mL were mixed and pH was 7.0 to 7.2. The mixture was sterilized at 121 $^\circ\text{C}$ for 20 min. The strains were inoculated on the slant of beef extract peptone medium and cultured at $(37 \pm 1)^\circ\text{C}$ for 24 h, and then stored at 0–5 $^\circ\text{C}$ (no greater than 1 month) as slant preservation bacteria. Slant preservation bacteria of *S. aureus* and *E. coli* were transferred to a solid culture medium of beef extract, cultivated for 24 h at $37 \pm 1^\circ\text{C}$ and a depurated single bacterial colony was obtained. The single colonies were picked and inoculated into a slant medium and activated, the fresh strain was activated after 2 consecutive transfers can be used in test. 4.5 mL of sterile liquid medium were added into the centrifuge tubes using a pipette. Fresh strains were scraped by inoculating loop from the slant culture medium to the sterile liquid culture medium and cultured for 24 h, and the sterile saline was added to the liquid nutrient medium to obtain the desired final concentration by the 10-fold serial dilution.

2.6. Determination of MIC of indole derivatives

The assays were performed using two-fold serial dilution tests to determine the MIC [27,28]. 2 mL of liquid medium and the test compounds were added to the test tube. The initial concentration of test compounds was 1 mg mL^{-1} , it was twice diluted in turns and then mixed. The test tube for inoculation of bacteria without the test compounds was used as positive control, and the test tube for non-inoculation of bacteria without the test compound was used as negative control, and chlorothalonil used as reference. All tubes were cultured in an incubator shaker for 24 h under 37 $^\circ\text{C}$. Via visual inspection, bacterial growth (turbid) was noted in the positive control tubes, by contrast, no bacterial growth (transparent) was noted in the negative control tubes. The absorbance values of the bacteria liquid in each test tube were measured in turn with a visible spectrophotometer. The standard curve of investigated sample was drawn with concentration as abscissa and absorbance as ordinate.

2.7. Algal inhibition test of indole derivatives

The algal solutions used during the exponential growth stage were diluted to various levels, and the absorbance values were measured at maximum absorption wavelength. The standard working curves for the algal solutions were calculated with various absorbance and concentration.

All experimental containers were handled with high-temperature sterilization techniques to ensure that no contaminants interfered with algal growth. The initial biomass of the algal solutions was uniformly diluted to about 1×10^5 algal cells per mL using f/2 medium and was

sufficiently low to allow exponential growth throughout the incubation period. The chlorothalonil used as reference. Acetone was used as a carrier to add indole derivatives (3 mg mL^{-1}) with low water solubility to the algal solutions, and the same quantity of acetone was added to all solutions, including the control groups. The test samples (0.5 mL) and diluted algal solutions (100 mL) were put in the glass vessels and transferred to the Versatile Environmental Test Chamber at a light intensity of 4000 lx and $20 \pm 2^\circ\text{C}$ with a 12: 12 h light: dark cycle. The absorbance values of algal solutions were tested every 24 h within the 7-day test period to obtain the algal growth status. The inhibition rates of algae were calculated using solutions from the seventh day as samples for the following formula:

$$T = (T_0 - T_i) / T_0 \times 100\%$$

where T is the inhibition rate of the algae, T_0 is the concentration of the control samples, and T_i is the concentration of the experimental samples. The measurements were repeated three times for each sample.

2.8. Anti-protein adsorption tests of marine antifouling paints

Coomassie brilliant blue method is commonly used to determine the BSA adsorption of marine antifouling paints [29,30]. Before this test, 0.5 mg mL^{-1} BSA solution was diluted to different concentrations with 0.15 M sodium chloride solution, pure sodium chloride solution was used as negative control, and the optical density at 595 nm ($\text{OD}_{595\text{nm}}$) was measured with UV-Visible-Near Infrared spectrophotometer (UV-3600 Plus, shimadzu, Japan). The standard curve of BSA solution was obtained by taking BSA concentration as the ordinate and $\text{OD}_{595\text{nm}}$ as the abscissa, and $\text{OD}_{595\text{nm}}$ value is the average value of the three measurements. The coated marine antifouling coating was put into 0.1 M PBS by ultrasonic for 20 min to make the surface clean, and then placed at 20 $^\circ\text{C}$ for 2 h. The pre-treated coatings were immersed into 10 mL 1 mg mL^{-1} BSA and incubated at 20 $^\circ\text{C}$ for certain hours. After BSA adsorption, the coatings were washed repeatedly by 10 mL 0.1 M PBS solution for several times to collect the adsorbed BSA. The obtained PBS solution with adsorbed BSA was dyed with Coomassie brilliant blue and detected at 595 nm, and unit area BSA adsorption amounts was calculated according to the standard curve. The experiment was repeated three times for each sample.

2.9. Marine field test

The antifouling property of paints were investigated according to the method for testing antifouling panels in shallow submergence (GB 5370–2007).

The low-carbon steel plates (250 mm $\times$ 150 mm $\times$ 3 mm) were polished and descaled with sandpaper (120 mesh), wiped with cotton yarn, and then brushed twice with epoxy primer. Moreover, the antifouling paints were brushed twice or the resins were brushed thrice. The plates were dried naturally in a well-ventilated place. Ultimately, the thickness of the dry film reached approximately 150 μm . The experimental plates painted with resins or antifouling paints were fixed in a slotted wooden frame, both ends of the wooden frame were fixed with iron bolts. The panels were immersed 1 to 1.5 m in the sea. Before immersed in the sea, the state of the sample frame such as the frame number, the sample number and the surface state were recorded. The samples should be fixed vertically and firmly on the frame when immersed in the sea. The surface of the sample plates was parallel to the main flow of seawater. Spacing between frames is not < 200 mm. The panels were regular inspected and taken photograph to record coating conditions. The antifouling performance of the coating was evaluated based on the type and amount of attached organisms on the sample plate, and the shedding state of the coating film.

3. Results and discussion

3.1. FTIR of 2a and 2b

FTIR spectra data of two compounds were listed in Table 2, their spectra are shown in Figs. S1 and S2. In both compound, the skeleton stretching vibrational absorption band of the aromatic ring at 1500 cm^{-1} to 1400 cm^{-1} were observed, and the stretching vibration band of the substituted aromatic ring at 700 cm^{-1} was appeared, indicating that both compounds contain Ring structure. Moreover, there was no peak of N-H stretching vibration in the vicinity of 3400 cm^{-1} , it can be seen that the N-position hydrogen on the indole ring has been replaced.

Two absorption bands of C = O = C at 1000 cm^{-1} to 1300 cm^{-1} were observed for compounds 2a and 2b, and the stretching vibration absorption peaks at 2920 cm^{-1} and 1470 cm^{-1} attributed to $-\text{CH}_3$ and $-\text{CH}_2$ respectively. It shows that the anthracycline has been attached to the ester group. Simultaneously, vibrational absorption peaks of N-H at 3200 cm^{-1} was appeared, and the carbonyl stretching vibration absorption peak at 1650 cm^{-1} corresponded to C = O, and the stretching vibration absorption peak of C-N at 1200 cm^{-1} was observed. It can be seen that the amide structure has been introduced into two compounds.

In the two compounds, stretching vibration peaks of the C = C at 1680 cm^{-1} to 1720 cm^{-1} were observed, and the stretching vibration and deformation vibration of = C-H at 3080 cm^{-1} and 920 cm^{-1} were observed, respectively. It indicates that the double bond has been introduced into two compounds.

3.2. ^1H NMR of 2a and 2b

2a- ^1H NMR(CDCl_3 , 600 MHz) δ : 2.78 (t, J = 6.9 Hz, 2H, CH_2CO), 4.12(m, 3H, CH_3), 4.39(t, J = 6.6 Hz, 2H, NCH_2), 4.61(d, J = 6.6 Hz, J = 5.4 Hz, 2H, CH_2NH), 5.64(d, J = 1.2 Hz, 1H, CH = CHH), 5.66(s, 1H, NH), 6.08(d, J = 10.2 Hz, 1H, CH = CHH), 6.31(d, J = 16.8 Hz, 1H,CH), 7.14 ~ 7.72(m, 4H, ArH).

2b- ^1H NMR(CDCl_3 , 600 MHz) δ : 1.20(t, J = 4.2 Hz, 3H, CH_3), 2.78 (t, J = 6.9 Hz, 2H, CH_2CO), 4.11(m, 2H, CH_2CH_3), 4.37(s, 2H, NCH_2), 4.58(d, J = 6.6 Hz, J = 5.4 Hz, 2H, CH_2NH), 5.62(d, J = 1.2 Hz, 1H, CH = CHH), 5.64(s, 1H, NH), 6.07(d, J = 10.2 Hz, 1H, CH = CHH), 6.30(d, J = 16.2 Hz, 1H,CH), 7.12 ~ 7.72(m, 4H, ArH).

^1H NMR spectra of the two synthesized products are shown in Figs. S3 and S4. From the data, it can be seen that the proton absorption peaks of fluorene ring at 7.12–7.72 were observed, indicating that the compound has a fluorene ring structure. Proton absorption peak at 4.40 attributed to $-\text{CH}_2$ of pyrrole ring, the absorption peak of N-H at 5.65 was observed, and the proton absorption peaks of = C-H at 5.60 to 6.30 were appeared, it indicates that the acrylamide group has been introduced into the 3-position of the indole ring. The absorption peak of $-\text{CH}_2$ attached to the nitrogen of the indole ring at 4.50 was observed, the proton absorption peak of $-\text{CH}_3$ or $-\text{CH}_2$ attached to ester group at 4.11 was appeared, indicating that propionate structure has been introduced into 1-position of indole ring. In conclusion, the chemical shift values, the number of protons, and the cleavage conditions of test compounds are completely consistent with the target compounds. Therefore, it can be confirmed that indole ester derivative containing acrylamide group has been synthesized.

Table 2
Infrared absorption spectra of compounds (unit: cm^{-1}).

compounds	$\nu_{\text{N-H}}$	$\nu_{\text{C=O}}$	$\nu_{\text{C-N}}$	$\nu_{\text{C-O-C}}$	$\nu_{\text{C=C}}$	$\nu_{\text{C=Aromatic skeleton}}$	$\nu_{\text{C-H}}$
2a	3281.27	1650.66	1210.55	1194.72 1318.21	1720.32	1536.68 1381.53 700.79	2917.15 1473.35
2b	3278.10	1650.66	1207.39	1188.39 1315.04	1717.15	1539.84 1381.53 697.63	2980.47 1473.35

3.3. Optimization of synthesis conditions of 2a and 2b

In the course of the Friedel-Crafts reaction, the influences of the catalyst type, reaction time and reaction temperature on the products were investigated.

3.3.1. Exploration of catalyst types

The catalyst has a significant effect on reaction yield in the Friedel-Crafts reaction. In order to select the appropriate catalyst, concentrated sulfuric acid, glacial acetic acid, anhydrous zinc chloride and anhydrous aluminum trichloride were investigated with other conditions keeping constant. The results are showed in Table 3.

It can be seen from Table 3 that four catalysts can all carry forward the reaction, but the yields of products are obviously different. Among them, catalytic yields of products are higher when catalysts are concentrated H_2SO_4 and anhydrous AlCl_3 , but the product obtained from concentrated H_2SO_4 catalyzed system is very viscous, which increases the difficulty of post-treatment and separation. Considering comprehensively, anhydrous AlCl_3 was the best catalyst in the reaction.

3.3.2. Exploration of reaction time

The Friedel-Crafts reaction is a reversible reaction. After the reaction reaches equilibrium, increasing the reaction time has little effect on the yield and even increases the amount of by-products. Reaction yields at 1, 2, 3, 5, and 7 days were tracked by column chromatography under other conditions unchanged.

The effect of reaction time on product yield is showed in Table 4. The yield increased significantly with the prolongation of the reaction time for two reactions, and the yield decreased slightly after the yield reaching the highest on the third day. It is presumed that the reaction reaches equilibrium on the third day and the yield no longer increases. Therefore, the optimal reaction time for the two reactions is 3 days.

3.3.3. Exploration of reaction temperature

The Friedel-Crafts reaction is a typical electrophilic substitution reaction. Generally, the reaction speeds up when the temperature increases but the side reactions also increase accordingly. Consequently, the reaction temperature is also an important factor for the yield of products. In the absence of other conditions unchanged, the yield of each reaction was studied at reaction temperatures of 10 $^\circ\text{C}$, 15 $^\circ\text{C}$, 35 $^\circ\text{C}$, and 50 $^\circ\text{C}$. The results are showed in Table 5.

For both reactions, the product yield increased with increasing temperature and reached a maximum yield at 25 $^\circ\text{C}$. The product yield decreased slightly when the temperature continued to rise. Considering comprehensively, the optimal reaction temperature is 25 $^\circ\text{C}$ for both

Table 3
Influence of different catalysts on the yields of two targets compounds.

catalysts	Concentrated H_2SO_4	glacial acetic acid	anhydrous ZnCl_2	anhydrous AlCl_3
yield of 2a (%)	greater than35	<15	<10	66.5
yield of 2b (%)	greater than34	<10	<10	70.8

Table 4

Influence of reaction time on the yields of two targets compounds.

Reaction Time (day)	1	2	3	5	7
yield of 2a (%)	30.2	50.3	65.5	64.9	60.4
yield of 2b (%)	32.5	52.5	68.6	65.7	62.3

Table 5

Influence of temperature on the yields of two targets compounds.

Reaction Time(°C)	10	15	25	35	50
yield of 2a (%)	49.6	66.7	68.3	60.4	60.2
yield of 2b (%)	50.7	66.3	67.4	62.5	59.3

reactions.

3.4. Antibacterial activity of 2a and 2b

In the experiment, commercially available antifouling agent chlorothalonil was used as a contrast. Absorbance value curves of the tested compounds at different concentrations are showed in Fig. 1 and Fig. 2. The minimum compound concentration at an absorbance value of 0 corresponds to the MIC. The results are showed in Table 6.

From the results, it can be seen that MIC values of the two synthesized compounds was lower than the chlorothalonil. It demonstrated that two compounds can significantly inhibit the selected bacteria, and their antibacterial properties are superior to chlorothalonil.

3.5. Anti-algal activity of 2a and 2b

The wavelength scanning diagram in Fig. S5 shows that the maximum absorption wavelengths of *C. vulgaris*, *N. closterium* and *S. costatum* are 681 nm, 682 nm and 688 nm, respectively. The standard work curves indicate that there is a significant positive and linear correlation between the absorbance and the concentration of algae, and the results are presented in Fig. 3. Anti-algal results of indole derivatives shows in Fig. 4 revealed 2a and 2b exhibit good ability to inhibit algae. The inhibition rates of compound 2b against *N. closterium* are more than 90%, and the inhibition rates of it is higher than that of 2a for the other two kinds of algae. This indicates that 2b is superior to 2a in inhibiting algae performance. It can be seen that the compound containing ethyl ester functional structure in indole derivatives has better algal inhibiting performance than that containing methyl ester functional structure. For the three kinds of algae, the inhibition rates of 2a and 2b are all higher

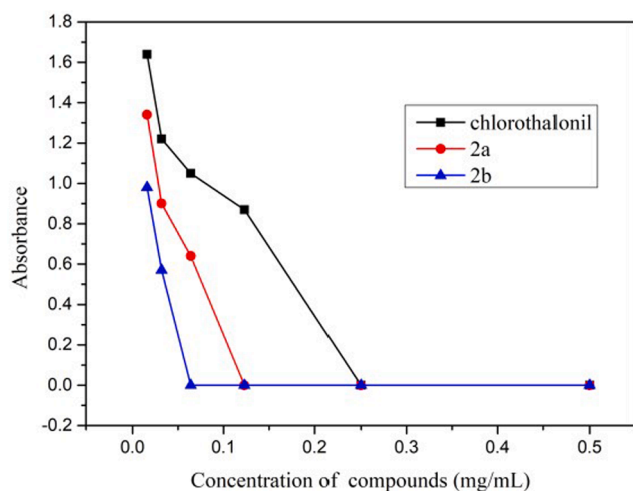


Fig. 1. Concentration dependent growth inhibiting activity of compounds against *E. coli*.

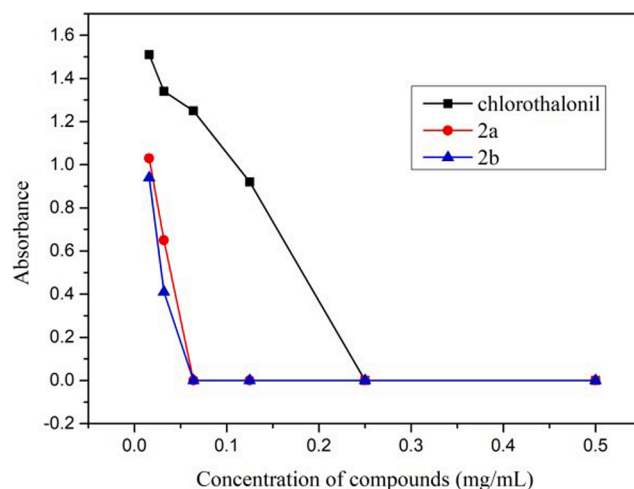


Fig. 2. Concentration dependent growth inhibiting activity of compounds against *S. aureus*.

Table 6

MIC (mg/mL) of compounds against selected bacteria.

Test organisms	Compounds		
	2a	2b	Chlorothalonil
<i>E. coli</i>	0.13	0.06	0.25
<i>S. aureus</i>	0.06	0.06	0.25

than that of chlorothalonil, which proves that the indole derivatives prepared have better algae inhibition performance. The inhibition rate of the compounds to the three kinds of algae was different, and the inhibition rate against *C. vulgaris* was the lowest.

The mechanism of algal inhibition is inferred that an indole derivative can act as an inhibitor of transmembrane transport and trigger algal cellular Ca^{2+} efflux, which reduces the abundance of cellular Ca^{2+} [31,32]. There are reports that calcium ions are relevant to the growth and development of algae [33,34] and they are involved in almost all physiological activities of algae [35,36]. Therefore, the Ca^{2+} efflux increased due to the presence of indole derivatives, which disrupted the Ca^{2+} balance in algal cells, then inhibited the growth of algae.

3.6. BSA adsorption activity of marine antifouling paints

The standard working curve of BSA solution in Fig. 5 shows a linear relationship between protein concentration and optical density.

Fig. 6 shows the amount of BSA adsorbed on antifouling paints immersed in 0.5 M BSA solution at different periods. With the increase of immersed time, the protein adsorption amount of the coating samples increased gradually. At each immersed time, the protein adsorption amount of antifouling paint with chlorothalonil as antifouling agent was the highest, and the protein adsorption amount of antifouling paint with 2b as antifouling agent was the least. It can be seen that the anti-protein adsorption property of indole derivatives is better than that of chlorothalonil, and the anti-protein adsorption property of 2b is better than that of 2a. Indole derivatives exhibit better protein resistance behaviors. The resistance to protein is propitious to inhibit the formation of fouling in marine environments, particularly at the early stage.

3.7. Marine field test of marine antifouling paints

Marine antifouling coatings containing two target products as antifouling agent were prepared. Chlorothalonil was used as performance references, and the widely used foreign antifouling paint P1 is used as reference for prepared paints. The antifouling activities of antifouling

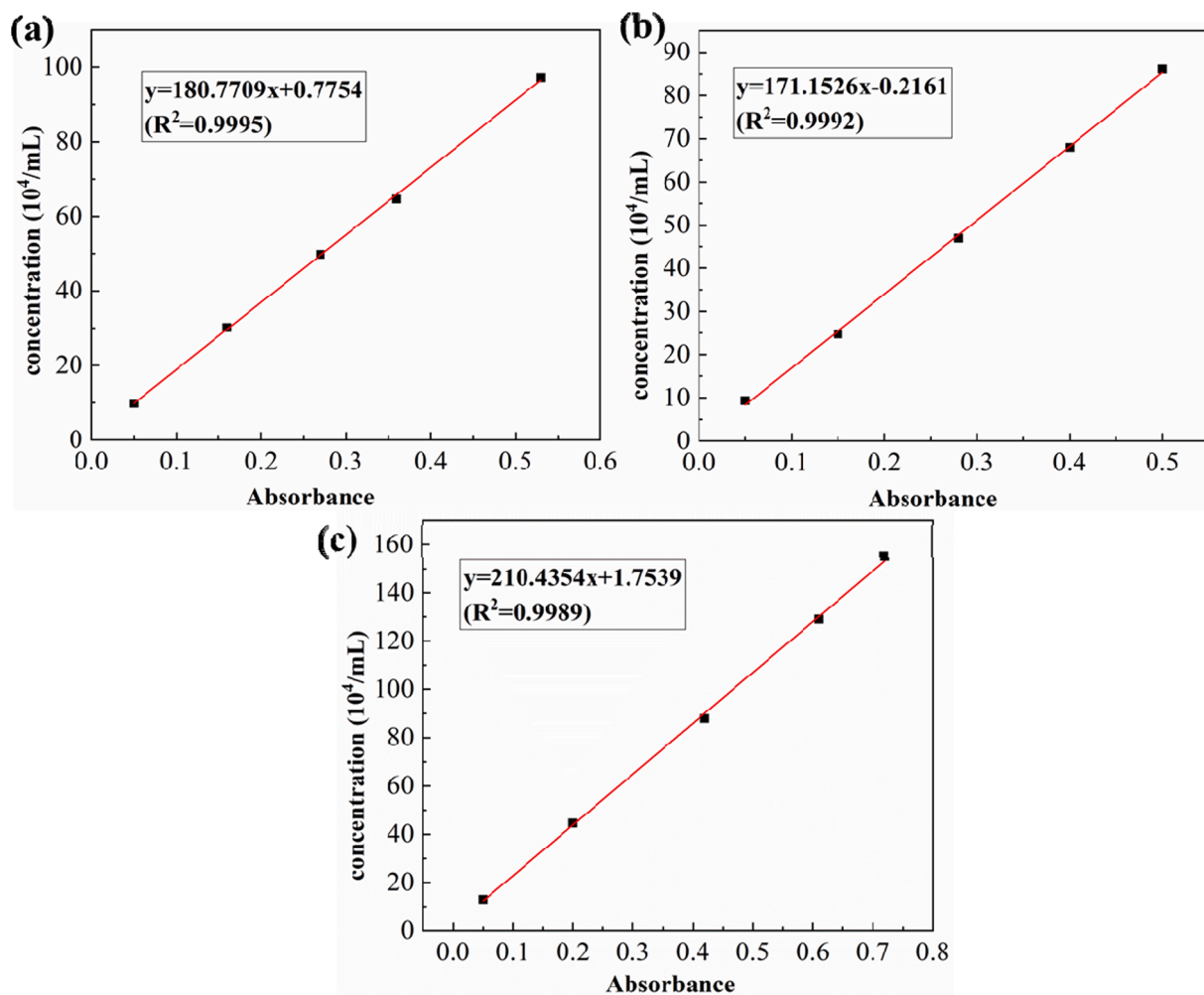
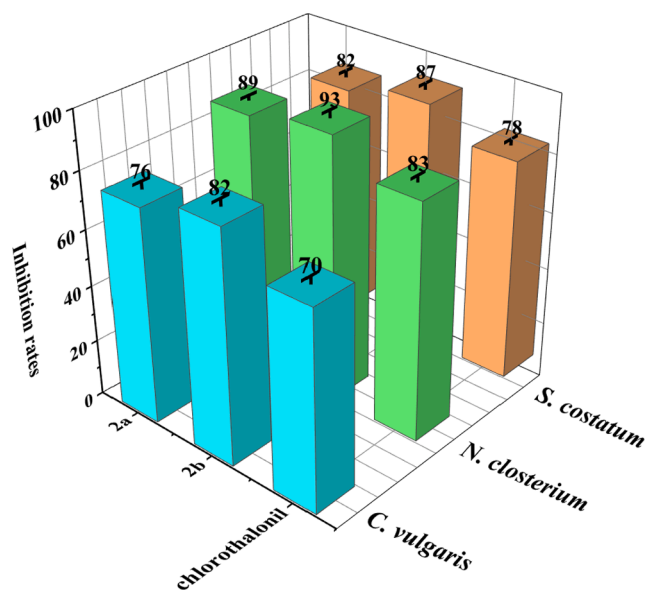
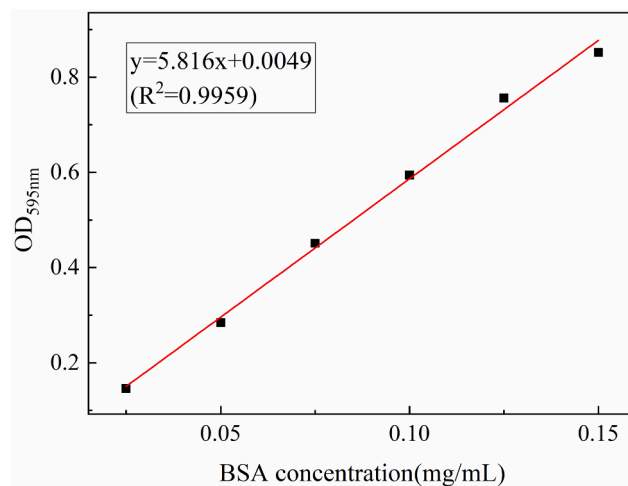
Fig. 3. Standard work curves of *C. vulgaris*(a), *N. closterium*(b) and *S. costatum*(c).Fig. 4. Anti-algal results of compounds against *C. vulgaris*, *N. closterium* and *S. costatum*.

Fig. 5. Standard work curves of BSA solution.

paints are showed in Table 7.

It can be seen from the pictures that the blank sample plate had been covered with biological mucous membranes and various bacteria after maintained for six months. The surface of the antifouling paints containing chlorothalonil and synthetic indole derivatives was very clean with no fouling adhered, so was the foreign antifouling paint P1. With

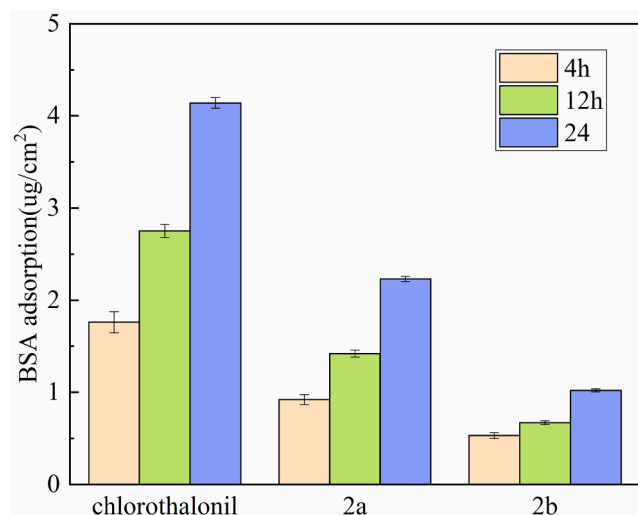


Fig. 6. BSA adsorption on marine antifouling paints.

Table 7
Antifouling activities of antifouling paints.

Antifouling agents	Time of immersing in the sea (M)			
	6	12	18	24
blank sample				
P1				
chlorothalonil				
2a				
2b				

the increasing of time, the blank plate was adhered by various organic organisms and gradually covered by large marine animals such as mussels and other organisms. Simultaneously, antifouling paint containing chlorothalonil was gradually attached by biological mucous membranes and bacteria even mollusks after 24 months. However, the surface of antifouling paints containing indole derivatives was always clean and only was attached by a small amount of biofilm. There is also only a small amount of sludge on the surface of the foreign antifouling paint P1. It can be concluded that the antifouling paints containing indole derivatives had better antifouling properties compared to the antifouling paint containing chlorothalonil. It can be proven that indole derivatives has better antifouling performance than chlorothalonil.

4. Conclusion

Two new indole derivatives containing acrylamide group were synthesized by Michael addition reaction and Friedel-Crafts reaction based on 5-bromo-1*H*-indole. The synthesized compounds were characterized by ¹H NMR and FTIR. In the step of the Friedel-Crafts reaction, the optimum process conditions were determined as that the catalyst was AlCl₃, the reaction time was 3 days and the reaction temperature was 25 °C. MIC of synthesized compounds and chlorothalonil against *S. aureus* and *E. coli* were investigated, the results showed that the two synthesized compounds have better antibacterial activity than chlorothalonil. Algal inhibiting experiment shows that indole derivatives have better performance algae inhibition performance. Anti-protein adsorption experiment of antifouling paints with indole derivative and chlorothalonil as antifouling agents proves that indole derivatives exhibit better protein resistance behaviors. There was no biological attachment on the surface of the antifouling paints using indole derivatives as antifouling agent after 24 months marine field test. All the experimental results show that indole derivatives have good inhibition of biological adhesion, and the antifouling paints with indole derivatives as antifouling agents have good antifouling properties.

CRediT authorship contribution statement

Chunhua Ni: Investigation, Writing – original draft. **Guobo Chen:** Validation. **Xia Li:** Conceptualization, Writing – review & editing. **Haizhou Zhao:** Resources. **Liangmin Yu:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Natural Science Foundation of China [grant number 41576077], nancial supports from Joint Funds of the Natural Science Foundation of China [grant number U1706225].

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cplett.2021.138994>.

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